

Project Title | Machine Learning–Based Anomaly Detection for Telecom Network Performance Data Ingestion Pipelines

1. Problem Statement

Technology Group of Dialog Axiata PLC relies on large-scale data pipelines to ingest daily network performance management (PM) data from multiple Network Management Systems (NMS) such as Huawei U2020, Ericsson ENM into centralized analytics platforms (e.g., Microsoft Fabric). These datasets are subsequently used for operational monitoring, capacity planning, fault and anomaly detection and strategic decision-making.

In practice, ingestion pipelines are vulnerable to multiple failure modes, including:

- connectivity outages between NMS and ingestion services
- partial or delayed file transfers
- schema mismatches
- duplicated records
- missing reporting intervals
- abnormal metric values caused by upstream system errors

When such failures occur, incomplete or corrupted datasets propagate into downstream analytical systems, producing misleading dashboards and inaccurate insights. These issues are typically discovered late, resulting in last-minute changes to the business decisions or engineering actions have already been planned based on the data.

Current detection approaches are often rule-based (e.g., threshold checks or row-count validation) or rely on manual inspection. Such methods are not stable and require constant maintenance as systems evolve, and it consumes unnecessary resources without generating much value to the business.

The aim of this capstone project is to design and implement an ML-driven anomaly detection system that operates at the data ingestion stage of a telecom performance data pipeline. The system will automatically analyse incoming daily batches of PM data, identify anomalous patterns that indicate ingestion or upstream failures, and flag them for corrective action such as re-triggering data collection jobs or raising operational alerts.

This project addresses a real-world industrial Data Science problem and focuses on building an end-to-end detection pipeline, from raw ingestion data to alerting and visualization.

2. Data Sources

2.1 Primary Dataset

The project will use multivariate telecom performance management time-series data, obtained from the organization with the respective PM counters.

Metrics are expected to include, throughput and traffic volume, cell availability, utilisation ratios, number of reporting network elements per interval etc

Each record will be timestamped and associated with a network entity such as a cell, site, or technology.

2.2 Simulated Ingestion Failures

To allow controlled evaluation, labelled anomalies representing pipeline failures will be injected into the dataset, such as, missing daily batches, partial ingestion of network elements, duplicated files, sudden volume drops and extreme spikes.

This will enable both supervised evaluation and experimentation with unsupervised or semi-supervised methods.

2.3 Data Characteristics

The dataset is structured as a multivariate time series and most of data are collected at an hourly granularity, allowing for detailed analysis of short-term patterns and temporal dynamics. In total, the dataset contains hundreds of thousands of records, making it large enough to support robust statistical modeling and machine-learning applications. This scale also implies the need for efficient storage, preprocessing, and computational techniques when performing analysis or forecasting tasks.

3. Intended Methods and Techniques

3.1 Problem Framing

- Problem Type: Anomaly / fault detection
- Technique Category: Machine Learning, statistical modelling, analytics
- System Context: End-to-end ingestion-stage detection pipeline with alerting

3.2 Baseline Approaches

Since the dataset has not been evaluated comprehensively, a concluded set of approaches are not available at this point. But it has been planned to implement simple statistical and heuristic checks such as:

- Z-score and IQR-based outlier detection
- rolling mean and variance deviation
- seasonal thresholding
- batch-volume consistency checks
- missing-interval detection

These provide transparent reference points that can be expanded to more advanced models that can be developed for further evaluations.

3.3 System Architecture

A prototype ingestion pipeline will be constructed with the following stages:

- a. Landing Zone: raw PM files arrive daily/hourly
- b. Pre-processing: cleaning, aggregation, feature engineering
- c. Anomaly Scoring: baseline + ML models
- d. Decision Layer: thresholds and ensemble logic
- e. Alerting: flagged batches written to logs / alerts

3.4 Link to Program Modules

The project draws directly on learning outcomes from several core subject areas, including Machine Learning, Big Data Analytics, Time Series Analysis, and Model Evaluation and Validation. These areas provide the theoretical foundation and practical tools needed to design, implement, and assess analytical models within the project. Where gaps in knowledge or technical depth are identified, further study will be undertaken to strengthen understanding and ensure rigorous evaluation. This continued learning will help improve the quality of the analysis, the reliability of results, and the overall robustness of the project's conclusions.

4. Expected Outputs

4.1 Technical Deliverables

Below listed technical deliverables are planned at the project completion stage.

- selected telecom PM dataset
- ingestion-stage feature engineering pipeline
- baseline statistical detectors
- trained ML anomaly detection model
- comparative experimental results
- reproducible code repository

4.2 Final Submission Items List

Below listed deliverables are planned to be submitted along with the final submission. Any further deliverables to be agreed with the supervisor during the upcoming feedback sessions.

- final technical report
- reproducibility checklist
- system documentation and trade-off analysis

5. Key Risks and Assumptions

The proposal has been drafted assuming below listed points and any deviation to be discussed with the supervisor and guidance on any required revisions to be obtained.

- PM data set can be accessed
- injected anomalies represent common ingestion failures
- pipeline can be prototyped within available compute resources
- partial labels are available for evaluation

5.1 Risks and Mitigation

The following risks are identified at the proposal stage of the project and any further identified will be accumulated along with the project progress

Risk	Impact	Mitigation
Lack of proper datasets	Reduced real world applicability	Got the consent from organization to assign me the project with the real data set to analyse
Overly complex datasets/anomalies	Model development risk	Identify sub datasets to perform ML/ shortlist a subset of anomalies
High false positives	alert fatigue	threshold tuning, ensembles
Scope creep	timeline risk	freeze dataset and architecture early

6. Summary

This project proposes a real world, industry-motivated data science solution that combines time-series anomaly detection with data-engineering practices to protect downstream analytics platforms from corrupted network performance data. By focusing on ingestion-stage detection, baseline-to-advanced model comparison, and reproducible evaluation, the project aims to deliver both technical depth and practical relevance expected through the capstone project.